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## Combustion and Flame

journal homepage: [www.elsevier.com/locate/combustflame](http://www.elsevier.com/locate/combustflame)Intermediate species measurement during *iso*-butanol auto-ignitionWeiqi Ji<sup>a</sup>, Peng Zhang<sup>a</sup>, Tanjin He<sup>b</sup>, Zhi Wang<sup>b</sup>, Ling Tao<sup>c</sup>, Xin He<sup>a,b,\*</sup>, Chung K. Law<sup>a,d</sup><sup>a</sup> Center for Combustion Energy, Tsinghua University, Beijing 100084, China<sup>b</sup> State Key Laboratory of Automotive Safety and Energy, Tsinghua University, Beijing 100084, China<sup>c</sup> National Renewable Energy Laboratory, Golden, CO 80401, USA<sup>d</sup> Department of Mechanical and Aerospace Engineering, Princeton University, Princeton, NJ 08544, USA

## ARTICLE INFO

## Article history:

Received 15 December 2014

Revised 11 June 2015

Accepted 12 June 2015

Available online xxx

## Key:

*Iso*-butanol

Rapid compression machine

Fast sampling

Ignition delay

Low temperature mechanism

## ABSTRACT

This work presents the time histories of intermediate species during the auto-ignition of *iso*-butanol at high pressure and intermediate temperature conditions obtained using a rapid compression machine and recently developed fast sampling system. *Iso*-butanol ignition delays were acquired for *iso*-butanol/O<sub>2</sub> mixture with an inert/O<sub>2</sub> ratio of 7.26, equivalence ratio of 0.4, in the temperature range of 840–950 K and at pressure of 25 bar. Fast sampling and gas chromatography were used to acquire and quantify the intermediate species during the ignition delay of the same mixture at  $P = 25.3$  bar and  $T = 905$  K. The ignition delay times and quantitative measurements of the mole fraction time histories of methane, ethene, propene, *iso*-butene, *iso*-butyraldehyde, *iso*-butanol, and carbon monoxide were compared with predictions from the detailed mechanisms developed by Sarathy et al., Merchant et al., and Cai et al. It is shown that while the Sarathy mechanism well predicts the overall ignition delay time, it overpredicts ethene by a factor of 6–10, underpredicts *iso*-butene by a factor of 2, and overpredicts *iso*-butyraldehyde by a factor of 2.

Reaction path and sensitivity analyses were carried out to identify the reactions responsible for the observed inadequacy. The rates of *iso*-butanol hydrogen atom abstraction by OH radical and the beta-scission reactions of hydroxybutyl radicals were updated based on recently published quantum calculation results. Significant improvements were achieved in predicting ignition delay at high pressures (25 and 30 bar) and the species concentrations of ethene and *iso*-butene. However, the updated mechanism still overpredicts *iso*-butyraldehyde concentrations. Also, the updated mechanism degrades the prediction in ignition delay at lower pressure (15 bar) compared to the original mechanism developed by Sarathy et al.

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## 1. Introduction

In recent years, the bio-butanol isomers including *n*-butanol, 2-butanol and *iso*-butanol have received extensive attention as the next generation biofuel [1]. This work focuses on *iso*-butanol because it is used as a blending component in petroleum fuels [2], while its oxidation mechanism is also a building block in developing the combustion kinetics of the larger alcohols.

Previous experiments on *iso*-butanol have been performed in flames [3–7], jet stirred reactors (JSR) [8], and shock tubes [9–15], providing kinetics targets for the development of *iso*-butanol kinetics mechanisms [8–10, 16–19]. Experiments have also been performed in rapid compression machines, although the acquired ignition delay times [20] exhibited significant discrepancies with model

predictions, especially at high pressure (30 bar) and low to intermediate temperature (754–857 K) conditions. Furthermore, intermediate species measurements were shown to be sensitive for evaluating the performance of *iso*-butanol kinetic mechanisms [8,12,19,21,22]. Specifically, from shock tube experiments Yasunaga et al. [12] measured the species concentration profiles during pyrolysis at pressures of 1.2–2.2 atm and temperatures of 1000–1500 K using gas chromatography (GC), while Stranic et al. [21] measured the species time histories of hydroxide (OH), water (H<sub>2</sub>O), carbon monoxide (CO), and methane (CH<sub>4</sub>) using UV and IR laser absorption during pyrolysis at pressure of 1.7 atm and temperatures of 1400–1520 K. These speciation studies provided insight into the high temperature reaction network for *iso*-butanol pyrolysis, especially the subset of reactions involving fuel decomposition [21]. Furthermore, Welz et al. [22] detected both transient and stable species profiles at 550 K and 750 K at low pressures (4–5.1 Torr) using multiplexed time-resolved synchrotron photoionization mass spectrometry, and Togbé et al. [8] investigated the *iso*-butanol oxidation in a JSR over the temperature range of 770–1250 K at 10 atm and with equivalence ratios of 0.5–4.0. These speciation studies provided insight into the low temperature

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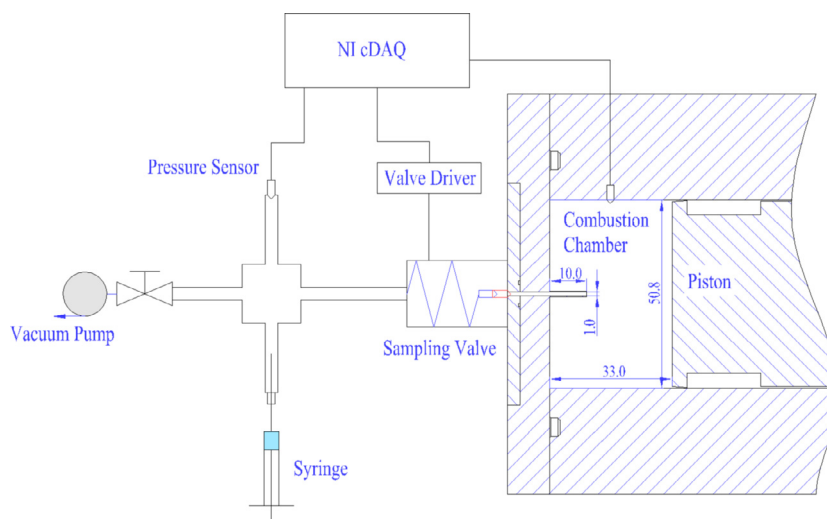


Fig. 1. The schematic of the fast sampling system.

reaction network for *iso*-butanol oxidation, especially the pathways involving oxygen addition to fuel radicals [22].

No speciation studies have been reported at high pressures and low to intermediate temperatures. Weber and Sung [20] commented that none of the existing models can predict the ignition delays at high pressure (30 bar) and low to intermediate temperature (754–857 K) well, and highlighted the importance of the peroxy chemistry in the low temperature pathways of *iso*-butanol oxidation based on sensitivity analysis of the ignition delay time using the Sarathy mechanism [17]. Without intermediate species information, the competing ratios between  $\beta$ -scission reactions and fuel propagation into the low temperature chain branching pathway of oxygen addition to fuel radicals were not investigated directly.

Recognizing the substantial deficiency of the *iso*-butanol chemistry, especially at intermediate temperature and high-pressure conditions, we have experimentally determined the time histories of the intermediate species of  $\text{CH}_4$ , ethene ( $\text{C}_2\text{H}_4$ ), propene ( $\text{C}_3\text{H}_6$ ), *iso*-butene ( $i\text{-C}_4\text{H}_8$ ), *iso*-butyraldehyde ( $i\text{-C}_3\text{H}_7\text{CHO}$ ), CO, carbon dioxide ( $\text{CO}_2$ ) and *iso*-butanol ( $i\text{-C}_4\text{H}_9\text{OH}$ ) during the auto-ignition of *iso*-butanol. Furthermore, the production pathways of the measured species were analyzed, species sensitivities to relevant chemical reactions based on the Sarathy mechanism [17] were discussed. The reaction rates of the key reactions, including *iso*-butanol hydrogen atom abstraction by OH radical and the beta-scission reactions of hydroxybutyl radicals, were updated based on recently published quantum calculation results. These then led to a substantially improved version of the Sarathy mechanism [17] to better predict *iso*-butanol auto-ignition. Details of this comprehensive investigation are presented herein.

## 2. Experimental method

### 2.1. The rapid compression machine

All experiments were performed using the Tsinghua University Rapid Compression Machine (TU-RCM). A detailed description of the TU-RCM can be found in Di et al. [23]. The mixture in the test section can be compressed to high pressure and high temperature within 25–30 ms. The compression ratio can be controlled by adjusting the length of the test section, and thus achieving different end-of-compression (EOC) gas temperature and pressure. A creviced piston design effectively captures the cooler gas in the boundary layer that is scraped from the chamber wall during compression, thus reducing its mixing with the hot gas in the center of the test section. The test

section was instrumented with a piezoelectric pressure transducer (KISTLER6125C) combined with a charge amplifier (KISTLER 5018A) for pressure measurements.

In this study, the compression and test sections of the rapid compression machine, the fast sampling valve, the sampling chamber, the mixing tank, and all tubing were heated to about 60 °C.

### 2.2. Test mixtures

The test mixture was prepared in a stainless steel mixing tank at room temperature and then maintained at 60 °C. The mixture composition was determined using the partial pressures of the mixture components. Ultra-high purity grade nitrogen ( $\text{N}_2 > 99.999\%$ ), argon ( $\text{Ar} > 99.999\%$ ), oxygen ( $\text{O}_2 > 99.995\%$ ), and *iso*-butanol ( $> 99.5\%$ , Sigma-Aldrich) were used. *iso*-butanol partial pressure was controlled below 60% of the saturated vapor pressure at room temperature to prevent condensation. All experiments were conducted at fuel-to-oxygen equivalence ratio of 0.4 with dilution ratio of 7.26, which was selected to avoid high pressure rise during ignition that could cause potential leakage through the sampling valve.

### 2.3. Fast sampling system

A fast sampling system was developed for intermediate species sampling and quantification. When conducting sampling experiments, a small amount of mixture was withdrawn from the core region of the combustion chamber during the ignition delay period. Only one fast sampling was performed for each ignition experiment. Then the sample was analyzed using gas chromatography (GC) for the species concentrations. By conducting a series of fast sampling experiments at different times during the ignition delay targeting at the same end of compression (EOC) condition, the time-histories of the intermediates were obtained.

The fast sampling system, shown in Fig. 1, mainly consists of a sampling probe, a fast-acting solenoid valve (Parker Series 9), a vacuum pump and a sample tank ( $\sim 5.4 \pm 0.1$  mL). The sample tank was equipped with a piezoresistive pressure transducer (KISTLER 4045A5) with an amplifier (KISTLER 4618 A0) and a septum port for gas extraction. Figure 1 shows the cross section view of the combustion chamber along with the sampling probe when the piston is stopped after compression. The clearance height of the test section is 33 mm. The sampling probe, mounted at the center of the end wall, is made of stainless steel with 1 mm in inner diameter. The opening of the sampling probe is located on the chamber center line and

10 mm away from the end wall. The other end of the probe is connected to the fast acting solenoid valve. A valve driver from Parker Inc. (IOTA ONE) is used to drive the fast sampling valve. During the experiment, the NI data acquisition and control system monitors the combustion chamber pressure. When the cylinder pressure reaches the threshold pressure (4 bar), the NI control system generates a TTL pulse signal, which is used as a trigger signal for the Parker IOTA ONE valve driver. For the current experiment, the duration of the valve driver is set at 1.5 ms. The time delay between the trigger signal and the valve driver signal is adjusted depending on the sampling time. Due to the excellent repeatability of the combustion chamber pressure trace, the sampling time with respect to the end of compression can be precisely controlled with run-to-run variation within 1 ms. A similar sampling apparatus was used in [24], but the new fast sampling system can work at higher pressures (more than 40 bar) to investigate the intermediates during the auto-ignition of *iso*-butanol at high pressures.

#### 2.4. Gas chromatography

A commercial GC (Agilent 7890B) equipped with three sets of columns was calibrated for quantitative measurements of species of interest (see Table S1 in the Supplemental Material for details). Temperature-controlled gas sampling valves inject the samples into the columns. A 250  $\mu$ L sample loop was adopted to quantify permanent gases using thermal conductivity detector (TCD), and a 25  $\mu$ L sample loop was adopted to quantify hydrocarbons and oxygenates using flame ionization detector (FID). Ultra high purity helium (Beijing Hua Yuan Gas, Purity Plus, 99.999%) was used as carrier gas for all detectors. Each FID used a hydrogen/air flame maintained at 300  $^{\circ}$ C. The TCD was set to 250  $^{\circ}$ C. Standard calibration mixtures of  $\text{CH}_4$ ,  $\text{C}_2\text{H}_4$ ,  $\text{C}_3\text{H}_6$ , *i*- $\text{C}_4\text{H}_8$ , CO and  $\text{CO}_2$  (Beijing Hua Yuan Gas) were used, while calibration mixtures of acetaldehyde ( $\text{CH}_3\text{CHO}$ ), propionaldehyde ( $\text{C}_2\text{H}_5\text{CHO}$ ), *iso*-butyraldehyde (*i*- $\text{C}_3\text{H}_7\text{CHO}$ ), and *iso*-butanol were prepared using the mixing tank. The retention times of acetone ( $\text{CH}_3\text{COCH}_3$ ), 2-propanol ( $\text{C}_2\text{H}_5\text{CH}_2\text{OH}$ ), *iso*-butenal (*i*- $\text{C}_3\text{H}_5\text{CHO}$ ) and *iso*-butanol (*i*- $\text{C}_3\text{H}_7\text{CH}_2\text{OH}$ ) were identified using their purified reagent. Signals from the GC detectors were recorded and analyzed using Agilent Open LAB CDS. The ratio of peak area to mole fraction was used as calibration factor.

### 3. Computational methods

Computational simulations were performed using the open source code Cantera [25] and CHEMKIN-Pro [26], and assumed a closed 0-D homogeneous batch reactor. In this study, the mechanism developed by Sarathy et al. [17], Merchant et al. [18], and Cai et al. [19] were studied for comparison with the experimental results.

For the majority of the computational simulations for the ignition delay time validation, a simplified model assuming a constant-volume adiabatic process (Constant Volume Model) was used and the initial conditions were the effective pressure and temperature (defined in Section 4.1), and the initial test gas composition. Heat loss during the ignition delay was partially compensated because the effective pressure was obtained by averaging the portion of the pressure trace after compression but before ignition.

Regarding the computational simulations for the species time history validation, a different modeling approach that better simulates the combustion process in the RCM was used. It is well recognized that combustion in an RCM is not a perfectly adiabatic, constant-volume process. Reaction can occur during compression. Post-compression heat loss in the boundary region causes the pressure to decrease in the test section, leading to temperature decrease in the core region. The processes during an RCM ignition experiment can be described as an isentropic compression of the core gases (due to the piston motion) followed by post-compression isentropic

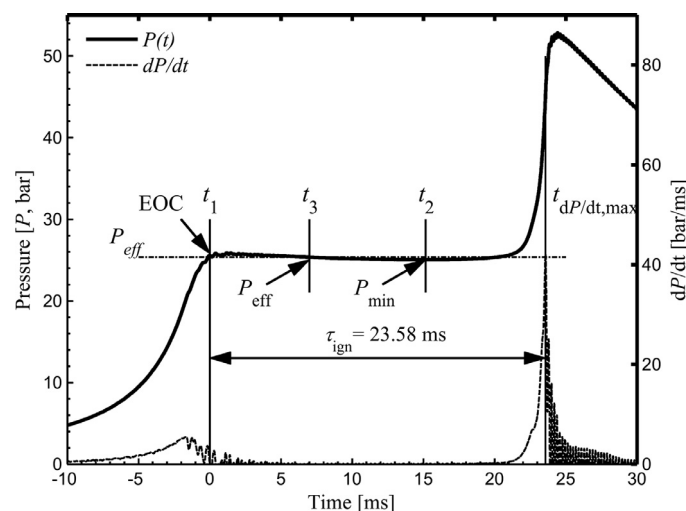


Fig. 2. The results for a typical *iso*-butanol ignition experiment. Condition:  $P_{\text{eff}} = 25.35$  bar,  $T_{\text{eff}} = 897$  K,  $\phi = 0.4$ , inert/ $\text{O}_2 = 7.26$ , and  $\tau_{\text{ign}} = 23.58$  ms.

expansion. In this work, the processes were represented as volume changes in the CHEMKIN calculations (Variable Volume Model). The volume time-history was obtained in the same manner as in He et al. [27]. The variable volume model was also used for the ignition delay time simulation using the Sarathy mechanism.

### 4. Experimental results

#### 4.1. Ignition experiments

Figure 2 presents the pressure trace of a typical TU-RCM *iso*-butanol ignition experiment. The original pressure data were filtered using a low-pass filter with cut-off frequency of 3000 Hz. A slight pressure oscillation was observed near the end of compression during the final stage of the piston deceleration. Based on the filtered pressure data, the pressure oscillation started at about 1 ms before the maximum pressure (due to compression), and the pressure difference between the first peak pressure in the oscillation and the maximum pressure is less than 4%. The uncertainty due to the compression process and the effects on the ignition delay time measurements were included in the uncertainty analysis.

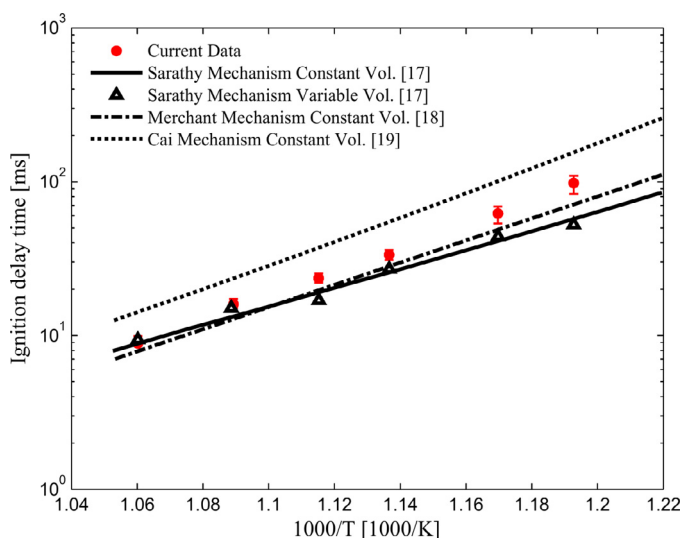
The end of compression was defined as the first pressure peak due to compression by the piston and this time was set as  $t = 0$  ms for all experiments. The local maximum of the derivative of the pressure with respect to time due to combustion is defined as the point of ignition. The ignition delay is then defined as the time interval between the end of compression and the start of ignition.

For each experiment, a non-igniting experiment with similar thermal properties as the igniting *iso*-butanol/ $\text{O}_2$ /inert gases mixture was also carried out by replacing  $\text{O}_2$  with  $\text{N}_2$  to characterize the heat losses after compression. The pressure history was then converted to the volume history to represent the compression and expansion processes of the core gas. The volume history, provided in the Supplemental Material, was used for the Chemkin simulation.

For each experiment, the effective test conditions are determined using the same method as previous TU-RCM experiments [23], where the effective pressure ( $P_{\text{eff}}$ ) was defined as the integrated average pressure from the first peak pressure (due to compression) to the minimum pressure ( $P_{\text{min}}$ ) after compression:

$$P_{\text{eff}} = \frac{1}{(t_2 - t_1)} \int_{t_1}^{t_2} P dt \quad (1)$$

where  $t_1$  and  $t_2$  are the time of the end of compression and the time of  $P_{\text{min}}$ , respectively. The key reference points for this calculation are

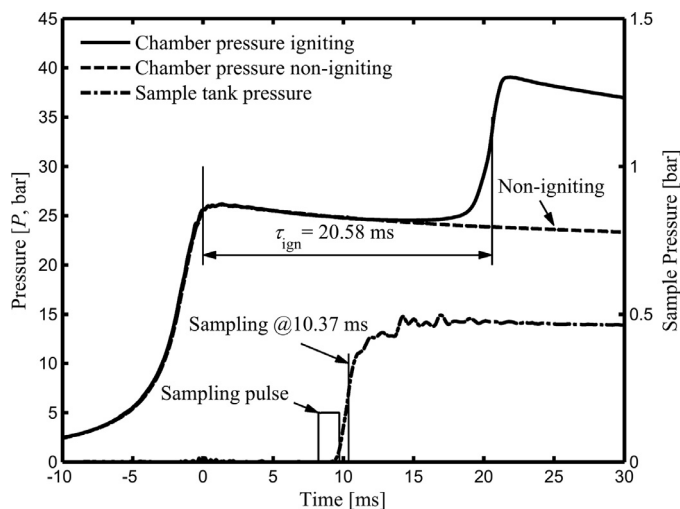


**Fig. 3.** Comparison of the predicted ignition delays and the RCM experimental results. The Constant Vol. simulations were done at condition of  $P = 25.2$  bar,  $\phi = 0.4$  and  $\text{inert}/\text{O}_2 = 7.26$ .

shown in Fig. 2. The effective temperature ( $T_{\text{eff}}$ ) was then calculated using the effective pressure ( $P_{\text{eff}}$ ), the initial pressure ( $P_0$ , the initial filling pressure before compression), and the initial temperature ( $T_0$ ) by numerically integrating the isentropic relation:

$$\int_{T_0}^{T_{\text{eff}}} \frac{\gamma}{\gamma - 1} d \ln T = \ln \left( \frac{P_{\text{eff}}}{P_0} \right) \quad (2)$$

where  $\gamma$  is the temperature dependent specific heat ratio of the test mixture, which was determined using the thermodynamics data of the Sarathy mechanism [17]. A summary of the experimental conditions and the measured ignition delay time ( $\tau_{\text{ign}}$ ), as well as the uncertainties of  $\tau_{\text{ign}}$  are provided in the Supplemental Material (Table S2). Both igniting experimental pressure traces and volume histories from non-igniting experiments are also included in the Supplemental Material. Figure 3 presents the experimental data from the current work and the model predictions based on the Sarathy mechanism [17], the Merchant mechanism [18], and the Cai mechanism [19]. Both constant volume model (Constant Vol.) and variable volume model (Variable Vol.) were used in the simulations using the Sarathy mechanism. The two modeling approaches yield very similar results for the ignition delay time, with a difference of less than 14% in the current work. Based on the good agreement between the two modeling approaches, the effective pressure and effective temperature defined herein represent well the post-compression experimental conditions and the constant volume model was considered adequate for evaluation of modeling trends. Consequently, for simplification only the constant volume model was used in simulations using the Merchant mechanism and the Cai mechanism. As is seen, the measured *iso*-butanol ignition delay time is quite linear on the Arrhenius scale, with no negative temperature coefficient behavior. Overall, both the Sarathy mechanism and the Merchant mechanism predict the ignition delay reasonably well over the entire temperature region examined, although the Sarathy mechanism underpredicts the ignition delay time by 20–40% at temperatures lower than 900 K. The reason for these discrepancies will be discussed along with the sampling results below. The Merchant mechanism underpredicts the ignition delay by about 30% over the entire temperature range studied. The Cai mechanism overpredicts the ignition delay by more than 50%.



**Fig. 4.** Results of a typical sampling experiment. Condition:  $P_{\text{eff}} = 25.1$  bar,  $T_{\text{eff}} = 905.8$  K.

#### 4.2. Fast sampling experiments

The ignition delay time study provided the basis to identify thermodynamic condition where fast sampling could be performed with good accuracy and repeatability. Based on the ignition delay experiments, EOC conditions of  $T_{\text{eff}} \approx 905$  K and  $P_{\text{eff}} \approx 25$  bar were selected for the fast sampling experiments. Experiments with  $T_{\text{eff}} = 905$  K yield an averaged ignition delay time of 21 ms, which allows a sampling time resolution of approximately 14:1. The typical sampling results, together with the time histories of the pressure in the combustion chamber and the sample tank, are presented in Fig. 4, where  $P_{\text{eff}} = 25.1$  bar,  $T_{\text{eff}} = 905$  K, and  $\tau_{\text{ign}} = 20.58$  ms.

As shown in Fig. 4, the gas sampling process does not cause noticeable change in the combustion chamber pressure because less than 0.4% of the gas in the combustion chamber was sampled. The pressure history in the sample tank increases rapidly during the sampling period of about 1.5 ms. The gas in the core region cooled down to about 275 K due to rapid expansion, which was later heated up to the sampling chamber temperature (333 K) or about 22% in temperature rise. This is the main reason of the slow pressure rise of about 20% after fast sampling. The pressure trace from non-igniting experiment is also shown in Fig. 4. For the period during the entire compression process, non-igniting pressure history is virtually identical to the igniting pressure history. Prediction using the Sarathy mechanism shows that less than 0.1% of *iso*-butanol is consumed at the end of compression, indicating no significant reaction during the compression process.

A summary of the pressure time histories of the gas sampling experiments is shown in Fig. 5. It is seen that while the data demonstrate good repeatability of the sampling experiments, slight differences in the EOC conditions and ignition delay times still exist. To compensate for the differences in experimental conditions, the sampling data are reported using a time domain that is normalized by the actual  $\tau_{\text{ign}}$  for each experiment. EOC is defined as  $t/\tau_{\text{ign}} = 0$ , and  $t/\tau_{\text{ign}} = 1$  corresponds to the time of ignition.

The mole fractions calculated from the gas chromatograms are not the species mole fractions of the gases in the core region of the RCM due to the existence of a dead volume in the sampling system. The dead volume ( $V_{\text{dead}}$ ) is defined as the total volume in the sampling probe and the fast sampling valve exposed to the reactant gases, which is estimated as 0.017 mL. During the compression process, the test gas mixture enters the sampling probe and the fast sampling valve from the core region. Because of the large surface-area-to-volume ratio, the gas temperature in the dead volume is low, and



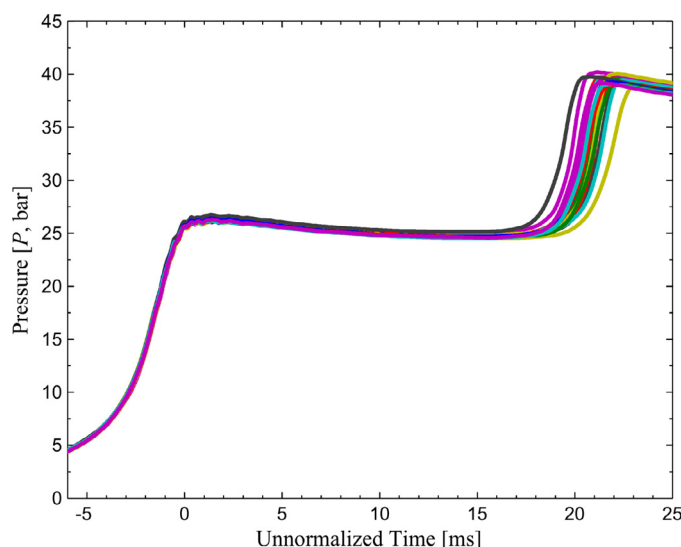


Fig. 5. Summary of the pressure time histories of the gas sampling experiments.

the gases are chemically quenched in this region. When the fast sampling valve is opened, these cold, unreacted gases are transported into the sampling tank, thereby diluting the gas sample obtained from the core region. The dilution ratio,  $\xi$ , is calculated using the following relation:

$$\xi = \frac{P_{\text{chamber}} \cdot V_{\text{dead}}}{T_{\text{dead}}} \cdot \frac{T_{\text{sampling}}}{P_{\text{sampling}} \cdot V_{\text{sampling}}} \quad (3)$$

where  $P_{\text{chamber}}$  is the average combustion chamber pressure during the sampling time,  $T_{\text{dead}}$  is the average gas temperature in the dead volume,  $P_{\text{sampling}}$  is the final pressure in the sample tank,  $V_{\text{sampling}}$  is the total volume of the sample tank, and  $T_{\text{sampling}}$  is the final gas temperature in the sample tank, which is 60 °C in the present study.

The species mole fraction in the core region,  $\chi_{\text{core}}$ , is given by

$$\chi_{\text{core}} * (1 - \xi) + \chi_{\text{dead}} * \xi = \chi_{\text{sampling}} \quad (4)$$

where  $\chi_{\text{sampling}}$  is the species mole fraction in the sample tank measured by GC.  $\chi_{\text{dead}}$  is the species mole fraction in the dead volume, and is estimated to be zero for the species other than *iso*-butanol. The mole fraction of *iso*-butanol in the core region was calculated assuming  $\chi_{\text{dead}}$  is equal to the *iso*-butanol mole fraction in the initial mixture (0.8%).

A summary of the experimental conditions, the measured  $\tau_{\text{ign}}$ , normalized sampling times, the upper and lower dilution ratio (defined in Section 4.3), the dilution corrected species concentrations, and the percentage of total carbon identified by GC measurement are given in the Supplemental Material (Table S3). Note that due to the low concentrations of intermediate species at the early stage of the ignition delay, majority of the sampling experiments were carried out at the second half of the ignition delay closer to the time of ignition. Figure 5 shows the pressure histories of the fast sampling experiments. The compression process and the slight decrease in pressure after EOC are consistent among different runs of the sampling experiments. The average  $P_{\text{eff}}$ ,  $T_{\text{eff}}$  and  $\tau_{\text{ign}}$  for the sampling experiments are  $25.3 \pm 0.1$  bar,  $905.1 \pm 1$  K and  $20.8 \pm 0.6$  ms, respectively.

Figure 6 shows typical GC-FID chromatograms for the experiment with normalized sampling time of 0.90. CO and CO<sub>2</sub> were measured using GC with TCD and are not shown here. Concentrations of the intermediate species for each sampling time were quantified using the calibration data for each species and the measured dilution ratio of gas sample. Acetone and 2-propenol were estimated using the calibration data of propionaldehyde. *iso*-butenal and *iso*-butenol were

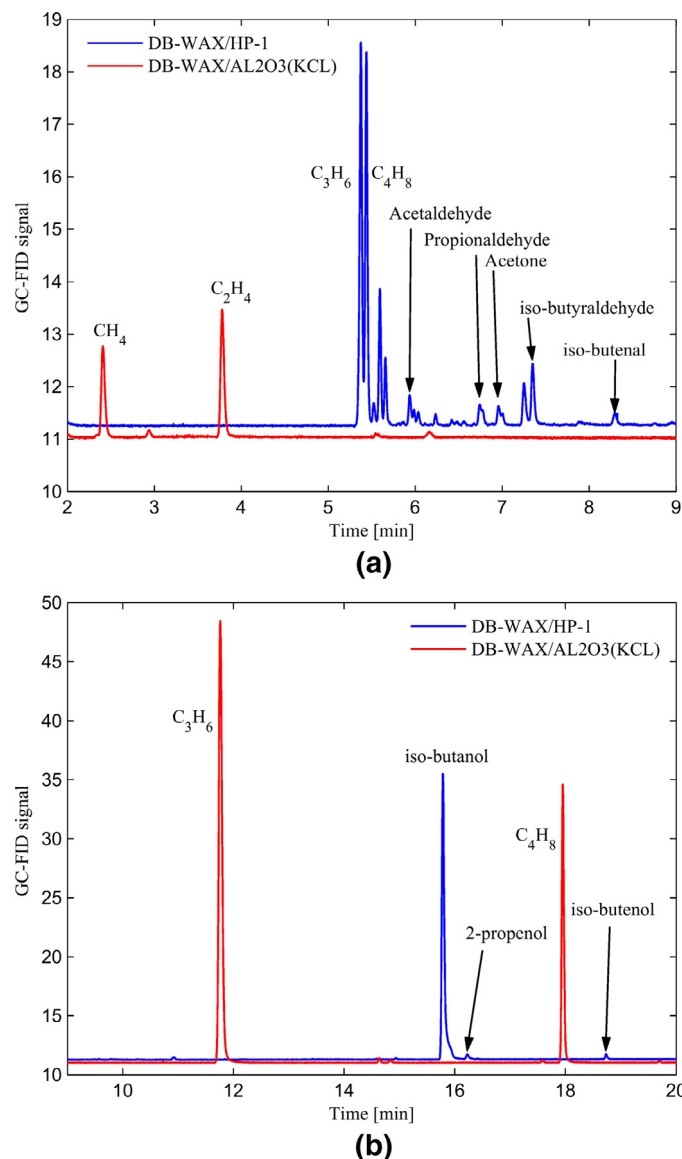


Fig. 6. Typical GC-FID chromatogram of a gas sample from an *iso*-butanol sampling experiment for the normalized sampling time of 0.90.

estimated using the calibration data of *iso*-butylaldehyde. The estimated concentrations of these species were listed in the Supplemental Material (Table S.3), without detailed comparison between the measured concentrations and the predictions by the mechanisms. The concentration of CO<sub>2</sub> is very low (<150 ppm) for NT<0.9, and CO<sub>2</sub> is also excluded from the detailed comparison. Besides, previous speciation studies in RCM [28] and JSR [8] have shown that the isomers of C<sub>2</sub>H<sub>4</sub>O may elute together with acetaldehyde, and the isomers of C<sub>3</sub>H<sub>6</sub>O may elute together with propionaldehyde. Thus acetaldehyde and propionaldehyde were excluded from the comparison. The concentrations of *iso*-butylaldehyde measured using FID were checked with MSD (Mass spectroscopy detector) and excellent consistency was observed, thus *iso*-butylaldehyde was included in the comparison. Several peaks corresponding to C<sub>2</sub> and C<sub>3</sub> oxygenates in the GC chromatogram were not identified. Besides, we were unable to measure the concentration of formaldehyde.

The experimental carbon balance ( $C_{\text{EXP}}$ ) for the GC quantified intermediate species, including CH<sub>4</sub>, C<sub>2</sub>H<sub>4</sub>, C<sub>3</sub>H<sub>6</sub>, *i*-C<sub>4</sub>H<sub>8</sub>, CH<sub>3</sub>CHO, C<sub>2</sub>H<sub>5</sub>CHO, CH<sub>3</sub>COCH<sub>3</sub>, *i*-C<sub>3</sub>H<sub>7</sub>CHO, *i*-C<sub>3</sub>H<sub>5</sub>CHO, 2-propenol, *iso*-butenol, CO<sub>2</sub>, CO and *iso*-butanol can be calculated using the

following equation:

$$C_{EXP} = \frac{\sum_i \chi_i * C_i}{\chi_{iso-butanol} * C_{iso-butanol}} \quad (5)$$

where  $\chi_i$  is the mole fraction of the  $i$ th species quantified in this study;  $C_i$  is the carbon number of the  $i$ th species;  $\chi_{iso-butanol}$  is the mole fraction of *iso*-butanol of the initial mixture; and  $C_{iso-butanol}$  is the carbon number of *iso*-butanol, which is equal to 4. The carbon balance predicted using the Sarathy mechanism ( $C_{CAL}$ ) was also included for comparison. The GC measurements account for 81–88% of the total carbon at early sampling times ( $t/\tau_{ign} < 0.5$ ), and 71–76% of the total carbon at sampling times closer to ignition ( $t/\tau_{ign} > 0.5$ ). The average ratio between the experimental carbon balance and the predicted carbon balance ( $C_{EXP}/C_{CAL}$ ) is about 84% at sampling times before ignition ( $t/\tau_{ign} \leq 0.95$ ). The experimental carbon balance is reasonably good and at level similar to the other speciation studies using RCM [28,29]. In this study, the carbon loss was primarily due to the *iso*-butanol condensation in the sampling probe. For experiments with sampling time very close to ignition ( $t/\tau_{ign} > 0.95$ ), most of *iso*-butanol has been consumed and most of carbon exists in species other than *iso*-butanol.

Figure 7 presents the results of eight species as a function of the normalized time for the nominal experimental condition of  $P_{eff, ave} = 25.3$  bar and  $T_{eff, ave} = 905$  K. The error bars are the uncertainties of the experimental data, described in detail below.

It is noted that radicals such as O, H, OH,  $HO_2$ , and  $CH_3$  can recombine to form water vapor and small hydrocarbons during gas sampling, which may interfere with the measurements of stable species. Concentrations of these small radicals predicted based on the Sarathy mechanism are presented in the Supplemental Material (Fig. S1). For the normalized time before 0.97, only  $HO_2$  exists in relatively high concentration ( $\sim 40$ – $100$  ppm). Prediction using the Sarathy mechanism shows no change in the quantified species concentrations during the rapid expansion and quenching.

### 4.3. Experimental uncertainties

#### 4.3.1. Uncertainties of the ignition delay measurement

Details about the uncertainty in ignition delay time can be found in [23], and the main conclusions are presented here. In this study, the effective pressure and temperature were used to represent the experimental conditions for the corresponding ignition delay time. In reality, pressure during the ignition delay is not constant, but decreases slowly. Thus, the upper and lower uncertainties of effective pressure are the integrated average pressure from  $t_1$  to the time where the pressure decreases to  $P_{eff}$  (see  $t_3$  in Fig. 2), and the integrated average pressure from  $t_3$  to  $t_2$  using the same approach as in Eq. (1). The other major pressure uncertainties are due to the uncertainty in the measurement of the initial pressure and the accuracy of the pressure transducer and charge amplifier, which is estimated to be  $\pm 0.5\%$ . The uncertainties in pressure are the direct sources of the corresponding uncertainties in the calculated effective temperature. The other major source of uncertainty in the temperature measurement is the uncertainty of the initial temperature ( $\pm 1$  K), which causes up to  $\pm 2.9$  K uncertainty in the calculated effective temperature. The uncertainty in the mixture composition is very small (less than 0.63%) and is thus neglected in this study. Another potential uncertainty is due to the reaction during compression. Combining the results of the uncertainty analysis, the uncertainty of the ignition delay time was determined for each experiment as:

$$\Delta\tau = \sqrt{\left(\frac{\partial\tau}{\partial P}dP\right)^2 + \left(\frac{\partial\tau}{\partial T}dT\right)^2 + (\Delta\tau_{compression})^2} \quad (6)$$

where  $\frac{\partial\tau}{\partial T}$  was calculated by taking the derivative of the Arrhenius fit of the experimental ignition delay time to temperature,

and  $\frac{\partial\tau}{\partial P}$  was estimated to be inversely related to pressure. The uncertainty due to the non-ideal compression process ( $\Delta\tau_{compression}$ ) was estimated based on the integral of the ignition delay time during compression, which is estimated to be 1 ms for the experiments conducted in this study. Note that the uncertainty due to the reaction during compression only increases the upper uncertainty limit.

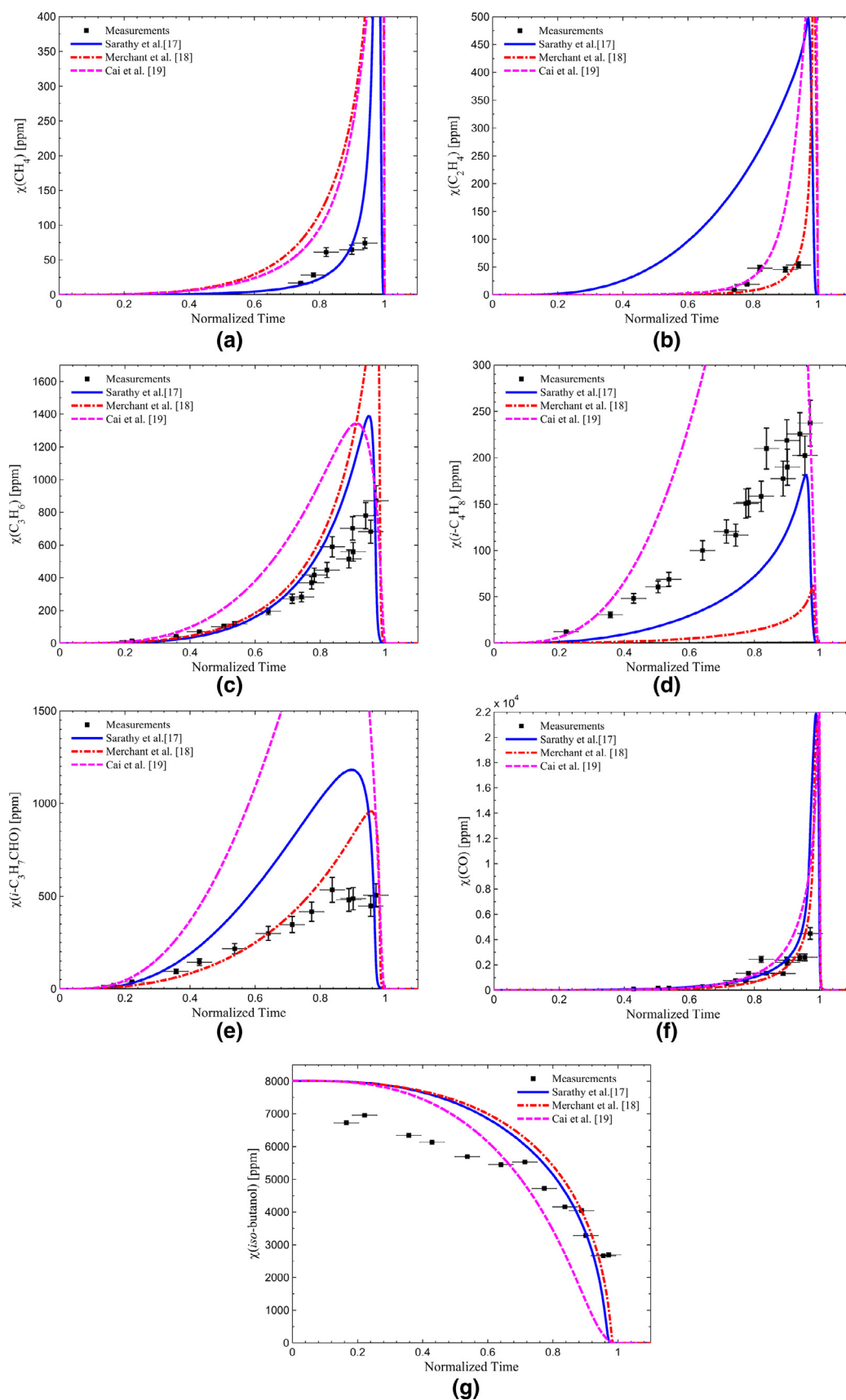
#### 4.3.2. Uncertainties of the species measurement

The primary sources of uncertainty in the fast sampling measurements are due to the dilution of the sample by the unreacted gases trapped in the ‘dead’ volume of the sampling probe. The gas temperature in the dead volume ( $T_{dead}$ ) cannot be directly measured and is the major source of the uncertainty in determining the dilution ratio. In this study, the average of the upper and lower limits of  $T_{dead}$  was used to calculate the dilution ratio, which is about 12%. The lower and upper limits of  $T_{dead}$  are the initial gas temperature ( $\sim 60^\circ\text{C}$ ) and the core gas temperature in the combustion chamber during sampling (which is very close to  $T_{eff}$  defined in Section 4.1). The upper and lower limits of the dilution ratio for each fast sampling experiment are presented in the Supplemental Material.

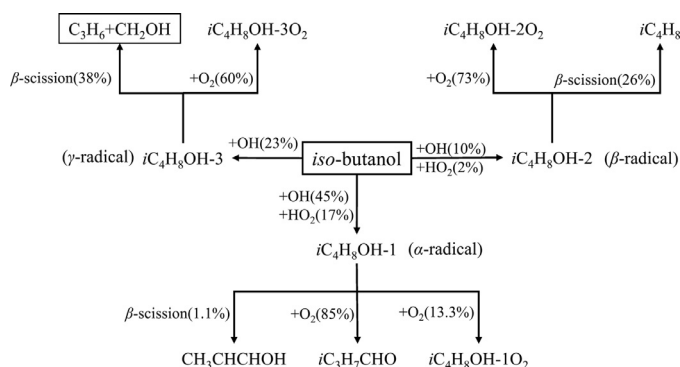
The uncertainty of  $\chi_{core}$  was calculated based on the uncertainty of  $\xi$  and the uncertainty of the GC measurements. The GC measurement uncertainty is estimated to be  $\pm 4\%$  of the species mole fraction for the permanent gases and hydrocarbons, and  $\pm 8\%$  for the oxygenates. For acetone, 2-propenol, *iso*-butenal, and *iso*-butanol, whose mole fractions were estimated based on the calibration data of other species, the GC measurement uncertainties are estimated to be  $\pm 20\%$ . The uncertainties of the dead volume dilution contribute a maximum error of  $\pm 8\%$  in the determination of the intermediate species mole fractions. The third uncertainty in the species measurements is due to variation in the experimental conditions and is estimated as a maximum of  $\pm 8\%$ . Combining the sources of uncertainty using the square root of the sum-of-the-squares approach yields an overall uncertainty in the species mole fraction of  $\pm 12\%$  for the directly calibrated permanent gases and hydrocarbons,  $\pm 14\%$  for the directly calibrated oxygenates (except *iso*-butanol), and  $\pm 23\%$  for the indirectly calibrated oxygenates.

It is difficult to quantify the uncertainties of *iso*-butanol due to the possibility of condensation in the dead volume. For the current study, the combustion chamber pressure after compression was about 25.3 bar, resulting in the partial pressure of *iso*-butanol of about 0.2 bar if no reaction occurs. This is higher than the saturated vapor pressure (0.12 bar) of *iso*-butanol at  $60^\circ\text{C}$ . Thus, the *iso*-butanol in the dead volume could condense on the inner surface of the sampling probe. When the gases in the core region flow through the sampling probe, condensation could also occur. As a consequence, the quantified mole fraction of *iso*-butanol tends to be lower than the actual mole fraction in the core region. Since it is not possible to estimate the actual amount of *iso*-butanol condensation, concentrations of *iso*-butanol were reported without error bars. Condensation could be avoided if the temperature of the test section could be heated to higher temperature. Unfortunately, the fast sampling valve tends to leak at temperature higher than  $60^\circ\text{C}$ , precluding fast sampling experiments at higher temperature. It is also worth to note that concentrations of all the other intermediate species are very low, and no condensation could occur.

The gas-sampling results presented in this study are the time averaged values of the species concentrations during the sampling time. Thus, the major source of uncertainties in time is due to the effects of finite sampling time. Since the sampling process spans 1.5 ms, the measured species mole fractions are necessarily time-averaged. This effect on the normalized time is represented as an uncertainty of half the sampling interval (approximately 0.75 ms), or  $\pm 0.0375$  in normalized time.



**Fig. 7.** Comparison between the measured intermediate species and the model predictions using variable volume model for the condition of  $P_{\text{eff}} = 25.3$  bar,  $T_{\text{eff}} = 905$  K,  $\chi(\text{iso-butanol}) = 0.8\%$ ,  $\chi(\text{O}_2) = 12.0\%$ ,  $\chi(\text{N}_2) = 55.9\%$ ,  $\chi(\text{Ar}) = 31.3\%$ .



**Fig. 8.** Reaction path diagram of the primary decomposition reactions for *iso*-butanol for  $\tau/\tau_{\text{ign}} = 0.75$  and the EOC condition of  $P = 25.3$  bar,  $T = 905$  K,  $\phi = 0.4$  and  $\text{inert}/\text{O}_2 = 7.26$ . Note the simulation was conducted using the constant volume model.

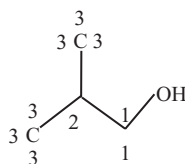
## 5. Model validation and optimization

### 5.1. Mechanism validation

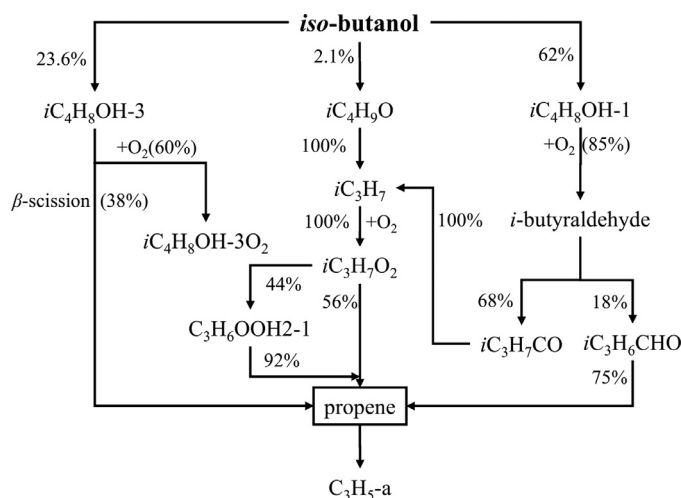
To take compression and post-compression heat loss into consideration, a variable volume model was utilized to predict the histories of the intermediate species. Figure 7 presents the model predictions using the variable volume model with the Sarathy mechanism, the Merchant mechanism, and the Cai mechanism for the condition of  $P_{\text{eff}} = 25.3$  bar,  $T_{\text{eff}} = 905$  K, with initial mole fractions of  $\chi(\text{iso-butanol}) = 0.8\%$ ,  $\chi(\text{O}_2) = 12.0\%$ ,  $\chi(\text{N}_2) = 55.9\%$ ,  $\chi(\text{Ar}) = 31.3\%$ . Overall, the Sarathy mechanism well predicts the evolution of  $\text{CH}_4$ ,  $\text{C}_3\text{H}_6$ , CO over the entire ignition delay period. However, it overpredicts  $\text{C}_2\text{H}_4$  by a factor of 6–10 and underpredicts  $i\text{-C}_4\text{H}_8$  by a factor of 2 over the entire ignition delay period, and overpredicts  $i\text{-C}_3\text{H}_7\text{CHO}$  significantly at the time close to ignition. The Merchant mechanism well predicts the evolution of  $\text{C}_2\text{H}_4$ ,  $\text{C}_3\text{H}_6$ ,  $i\text{-C}_3\text{H}_7\text{CHO}$  and CO. However, it overpredicts  $\text{CH}_4$  by a factor of 5 and underpredicts  $i\text{-C}_4\text{H}_8$  by a factor of 5 over the entire ignition time period. The Cai mechanism predicts a faster consumption of *iso*-butanol, thus it overpredicts  $\text{CH}_4$  by a factor of 5, overpredicts  $\text{C}_3\text{H}_6$  by a factor of 2, overpredicts  $i\text{-C}_4\text{H}_8$  by a factor of 2–3, and overpredicts  $i\text{-C}_3\text{H}_7\text{CHO}$  by a factor of 3–4 over the entire ignition delay period.

### 5.2. Reaction path and sensitivity analyses

For the discussions presented below, we are following the notation used by Sarathy et al. [17] for the molecular formulas, in which the last number following “-” represents the location of the unpaired site:



The rate of production analysis was conducted using the Sarathy mechanism to identify the reaction pathways important for the species measured. The analysis was conducted at 75% of  $\tau_{\text{ign}}$ , which corresponds to 20% of the fuel consumption based on the constant volume model. Figures 8–10 show the primary reaction pathways for *iso*-butanol and intermediates measured. Reaction pathways for  $\text{CH}_4$  and CO are not shown as the concentrations are very low at 75% of  $\tau_{\text{ign}}$  as indicated by the model predictions and experimental measurements. The percentages listed in the figures quantify the specific reaction pathway relative to the overall consumption of the



**Fig. 9.** Reaction path diagram of the primary formation pathways for propene. Same modeling condition as Fig. 8.

individual species. Figures 11–13 show the sensitivity analysis of three species, namely ethene, *iso*-butene and *iso*-butylaldehyde. The analysis was using a constant volume model, and the normalized sensitivity coefficient is defined as

$$S_{j,i} = \frac{\partial \chi_j}{\partial k_i} \cdot \frac{k_i}{(\chi_j)_{\text{local}}} \quad (7)$$

where  $\chi_j$  is the mole fraction of species  $j$  and  $k_i$  is the rate coefficient of reaction  $i$ .

Results for *iso*-butanol (Fig. 8) show that H-atom abstractions via the OH radical and  $\text{HO}_2$  radical are the dominant pathways for the *iso*-butanol decomposition. The subsequent reactions of oxygen addition to hydroxybutyl are comparable to the  $\beta$ -scission reactions for the  $\beta$ - and  $\gamma$ -branch at current conditions, while the direct formation of *iso*-butylaldehyde +  $\text{HO}_2$  from oxygen addition to  $\alpha$ -hydroxybutyl radical is dominant in the  $\alpha$ -branch. Figure 9 shows that propene is predominantly formed via the  $\beta$ -scission of  $i\text{-C}_4\text{H}_8\text{OH-3}$ , decomposition of  $i\text{-C}_3\text{H}_6\text{CHO}$ , and  $i\text{-C}_3\text{H}_7 + \text{O}_2 = i\text{-C}_3\text{H}_7\text{O}_2$  with the subsequent decomposition as well as isomerization reactions.  $i\text{-C}_4\text{H}_8\text{OH-3}$  is formed via the  $\gamma$ -branch of H-atom abstractions from *iso*-butanol.  $i\text{-C}_3\text{H}_7\text{CO}$  (precursor of  $i\text{-C}_3\text{H}_7$ ) and  $i\text{-C}_3\text{H}_6\text{CHO}$  are formed due to the consumption of *iso*-butylaldehyde, which is formed via the reaction  $i\text{-C}_4\text{H}_8\text{OH-1} + \text{O}_2$ .

The largest discrepancies (Fig. 7) between the model predictions and experimental results are the ethene concentrations, in which the model overpredicts by a factor of 6–10 during the entire ignition delay period. Figure 10 shows that ethene is formed solely through the pathway initiated by H-atom abstraction of *iso*-butanol to form  $i\text{-C}_4\text{H}_8\text{OH-3}$  and followed by  $i\text{-C}_4\text{H}_8\text{OH-3} + \text{O}_2$  low temperature chain branching reactions. The rate of production analysis indicates that either the  $\gamma$ -branch of *iso*-butanol decomposition is too fast comparing to other branches or  $i\text{-C}_4\text{H}_8\text{OH-3} + \text{O}_2$  low temperature chain branching reaction is too fast comparing to the  $\beta$ -scission of  $i\text{-C}_4\text{H}_8\text{OH-3}$  radical. This is supported by the sensitivity analysis of the ethene species, as shown in Fig. 11. The ethene species concentration is very sensitive to the initial reaction of the *iso*-butanol + OH-abstraction to form  $i\text{-C}_4\text{H}_8\text{OH-3}$  and the reaction of  $\beta$ -scission of  $i\text{-C}_4\text{H}_8\text{OH-3}$  to form  $\text{CH}_2\text{OH}$  and propene. Additionally, the ethene species concentration is also sensitive to the *iso*-butanol + OH-abstraction to form  $i\text{-C}_4\text{H}_8\text{OH-1}$ , which competes with the formation of  $i\text{-C}_4\text{H}_8\text{OH-3}$ .

The model underpredicts *iso*-butene by a factor of 2 during the ignition delay. *Iso*-butene (Fig. 10) is primarily formed via the  $\beta$ -scission of  $i\text{-C}_4\text{H}_8\text{OH-2}$ , which is formed via the  $\beta$ -branch of H-atom abstractions of *iso*-butanol. *Iso*-butene is consumed by the



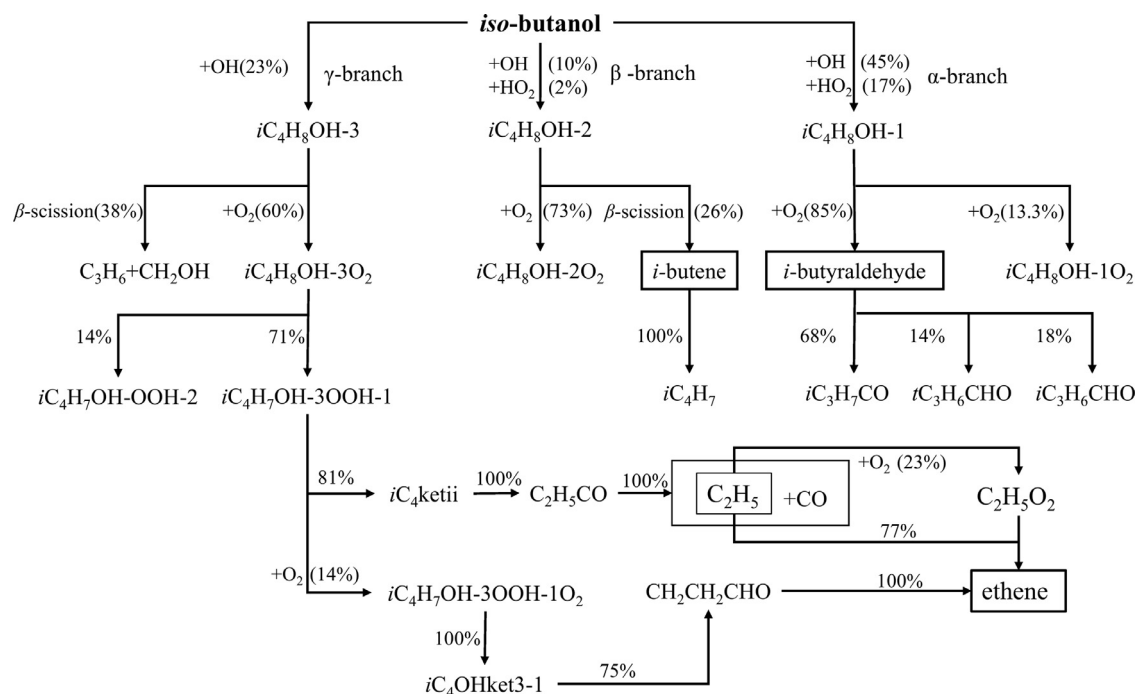


Fig. 10. Reaction path diagram of the primary formation pathways for ethene, iso-butene, and *i*-butyraldehyde. Same modeling condition as Fig. 8.

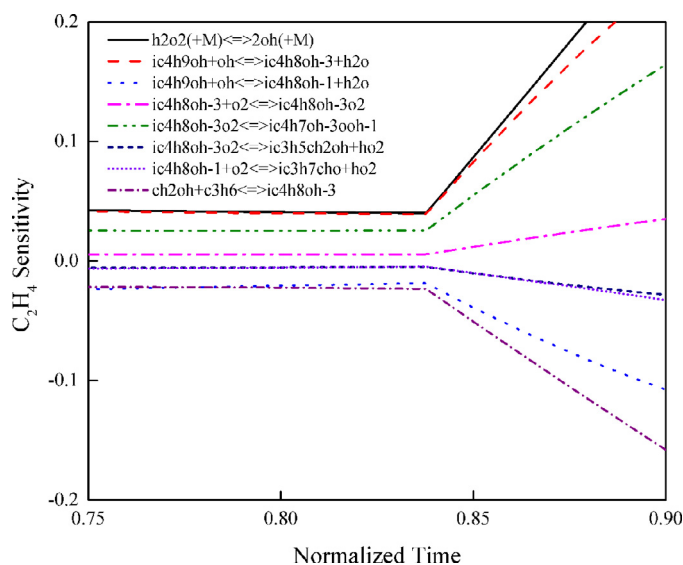


Fig. 11. Sensitivity of  $C_2H_4$ . Same modeling condition as Fig. 8.

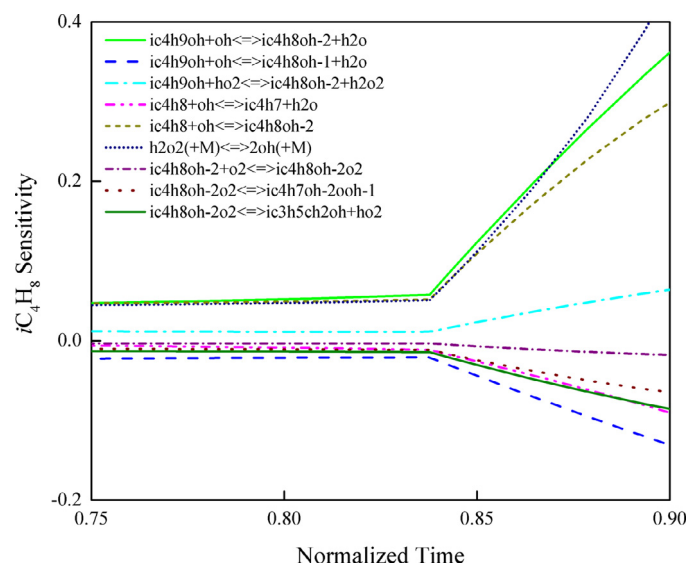


Fig. 12. Sensitivity of  $i-C_4H_8$ . Same modeling condition as Fig. 8.

H-atom abstraction to form the  $i-C_4H_7$  radical. The reaction path way analysis indicates that either the  $\beta$ -branch of *iso*-butanol decomposition is too fast or the  $\beta$ -scission of  $i-C_4H_8OH-2$  reaction is too slow. This conclusion is also supported by the sensitivity analysis of *iso*-butene species concentration (Fig. 12), showing that the most sensitive reactions are the *iso*-butanol + OH-abstractions to form the  $i-C_4H_8OH-2$  and  $i-C_4H_8OH-1$  isomers, as well as the  $\beta$ -scission of  $i-C_4H_8OH-2$  to form *iso*-butene.

The mechanism overpredicts *iso*-butyraldehyde by a factor of 2 at the time close to ignition.  $i-C_3H_7CHO$  (Fig. 10) is primarily formed via oxygen addition to the  $i-C_4H_8OH-1$  radical, which is formed via  $\alpha$ -branch of H-atom abstraction from *iso*-butanol. The reaction path analysis indicates that either the branching fraction for the  $i-C_4H_8OH-1$  channel of *iso*-butanol decomposition is too fast or the reaction  $i-C_4H_8OH-1 + O_2 = i-C_4H_8OH-1O_2$  is too slow comparing to

$i-C_4H_8OH-1 + O_2 = i-C_3H_7CHO + HO_2$  for the condition studied here. Besides, the over prediction of *iso*-butyraldehyde may be due to the consumption reactions of *iso*-butyraldehyde being too slow for the conditions studied. However, sensitivity analysis (Fig. 13) shows that  $i-C_3H_7CHO$  species concentration is not sensitive to  $i-C_4H_8OH-1 + O_2 = i-C_4H_8OH-1O_2$ .

### 5.3. Mechanism optimization

Since the major discrepancies between the model predictions and experimental results are the ethene concentrations, efforts were taken to optimize the rate constants of the key reactions leading to the formation of ethene.

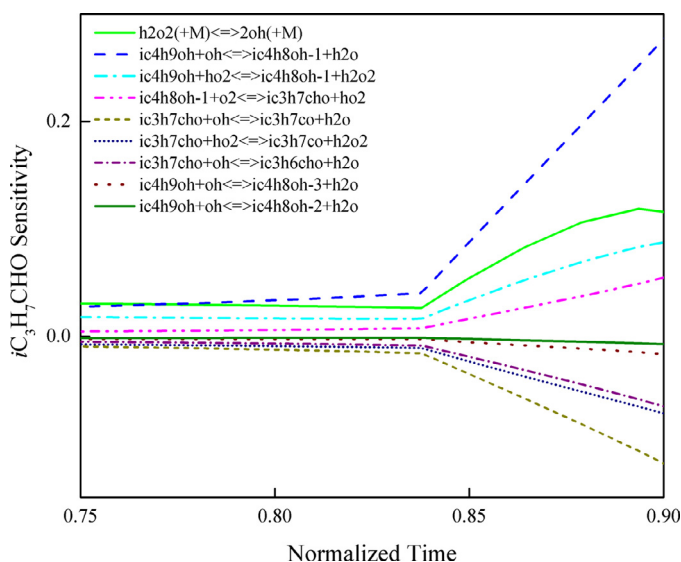


Fig. 13. Sensitivity of  $i\text{-C}_3\text{H}_7\text{CHO}$ . Same modeling condition as Fig. 8.

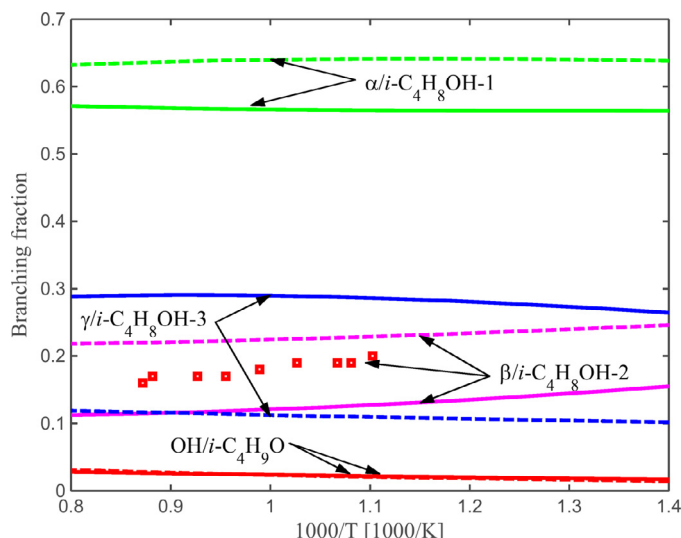


Fig. 15. Branching fractions for the four H-atom abstraction channels by OH from  $\alpha$ ,  $\beta$ ,  $\gamma$  carbon sites and from the alcohol group of *iso*-butanol. Solid lines: Sarathy mechanism [17], dashed lines: Zheng et al. [30], red square symbols: Pang et al. [31].

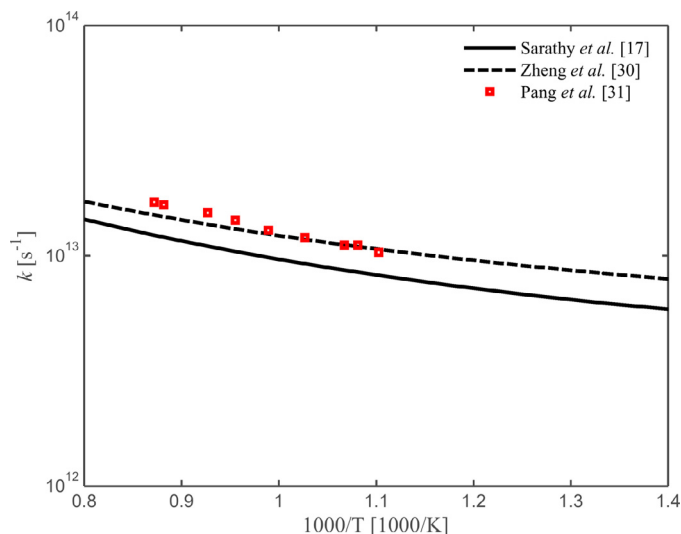


Fig. 14. Rate constants for the overall *iso*-butanol + OH reaction from Zheng et al. [30], Sarathy mechanism [17], and Pang et al. [31].

### 5.3.1. H-atom abstractions of *iso*-butanol by OH radical

The rate coefficients for the reactions of *iso*-butanol with OH have been the subject of recent experimental and theoretical studies [30,31]. Pang et al. [31] measured the rate constant of the non- $\beta$  channels in the temperature range of 907–1147 K at near atmospheric pressures using a shock tube, leading to the expression ( $k_{\text{overall}} - k_{\beta}$ ) =  $1.84 \times 10^{-10} \exp(-2350/T [\text{K}]) [\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}]$ . The shock tube data are in good agreement with the recent theoretical calculations of Zheng et al. [30], who used multi-path variational transition-state theory with small curvature tunneling (MP-CVT/SCT) and taking vibration anharmonicity into account, to determine the overall rate constants for *iso*-butanol + OH-abstractions. Results (Fig. 14) show that the overall rate constants of Zheng et al. [30] are consistent with Pang et al. [31], which are higher than that of the Sarathy mechanism by a factor of 1.3 at the temperature range of 700–1100 K.

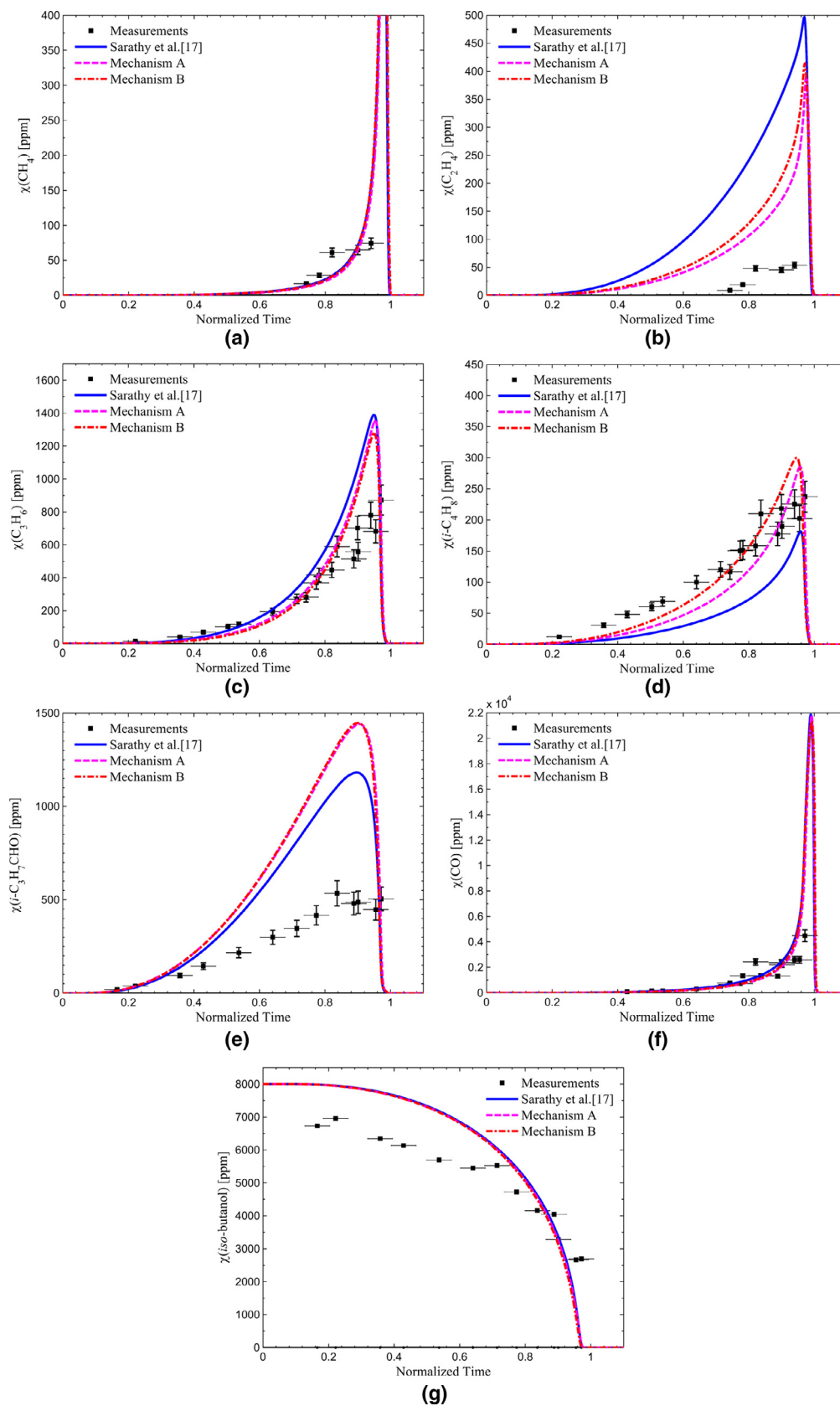
Figure 15 shows that the branching fractions for four H-atom abstraction branches by the OH radical from the  $\alpha$ ,  $\beta$ ,  $\gamma$  carbon sites and from the OH group of *iso*-butanol. The legends indicate the site of the H-atom abstraction and the corresponding radicals produced. The solid lines are the values used in the Sarathy mechanism, and

the dashed lines are those recommended by Zheng et al. [30] based on the MP-CVT/SCT( $P = 4$ ) calculation. The recommended branching fractions for the  $\beta$  carbon site from Pang et al. [31] at the temperature range of 900–1150 K (square symbols) are also plotted for comparison. It is clear that the branching fractions of  $\alpha$ -channel (producing  $i\text{-C}_4\text{H}_8\text{OH-1}$ ) and  $\beta$ -channel (producing  $i\text{-C}_4\text{H}_8\text{OH-2}$ ) recommended by Zheng et al. [30] are higher than those adopted in the Sarathy mechanism, while the branching fractions of  $\gamma$ -channel (producing  $i\text{-C}_4\text{H}_8\text{OH-3}$ ) are lower.

To quantify the effects of changing the overall rate coefficients and the branching fractions of *iso*-butanol + OH reactions, the recommended reaction rates from Zheng et al. [30] were used to modify the Sarathy mechanism. The modified reaction mechanism, named Mechanism A, leads to the change of the ignition delay time at  $P_{\text{eff}} = 25.3$  bar,  $T_{\text{eff}} = 905$  K, and  $\chi$  (*iso*-butanol) = 0.8%, from 17.5 to 19.7 ms. As expected based on the reaction path and sensitivity analyses presented above, the modified reaction mechanism predicts lower ethene concentrations, which better matches the measured ethene species profile as shown in Fig. 16. As the new rates lead to higher branch fractions of the  $\alpha$ -branch (producing  $i\text{-C}_4\text{H}_8\text{OH-1}$ ) and  $\beta$ -branch (producing  $i\text{-C}_4\text{H}_8\text{OH-2}$ ), Mechanism A predicts slightly higher  $i\text{-C}_4\text{H}_8$  concentrations and *iso*-butyraldehyde concentrations. Thus Mechanism A better matches the measured  $i\text{-C}_4\text{H}_8$  species profile, while slightly overpredicts the *iso*-butyraldehyde concentrations comparing to Sarathy mechanism. Mechanism A also slightly better predicts the propene concentrations.

### 5.3.2. $\beta$ -scission reactions of $i\text{C}_4\text{H}_8\text{OH}$ radicals

Mechanism A still overpredicts ethene by a factor of 5 during the ignition delay. The sensitivity analysis (Fig. 11) shows that the formation of ethene is also sensitive to the reaction of  $i\text{-C}_4\text{H}_8\text{OH-3} = \text{C}_3\text{H}_6 + \text{CH}_2\text{OH}$ , as it competes with the low temperature pathway of oxygen addition to hydroxybutyl radical leading to the formation of ethene. The rates of  $i\text{-C}_4\text{H}_8\text{OH-3} = \text{C}_3\text{H}_6 + \text{CH}_2\text{OH}$  were given in the reverse direction, namely  $\text{CH}_2\text{OH}$  radical addition to  $\text{C}_3\text{H}_6$ , in the Sarathy mechanism, and were estimated based on the reaction of  $\text{CH}_3$  radical addition to propene to form  $i\text{C}_4\text{H}_9$  radical, whose reaction rates were estimated by Curran [32]. In the recent mechanism developed by Cai et al. [19] the rates of  $\beta$ -scission reactions of  $i\text{C}_4\text{H}_8\text{OH}$  radicals were theoretically calculated with pressure- and temperature-dependence, and the rates were given in dissociation direction to eliminate the uncertainties raised by the inaccuracy in

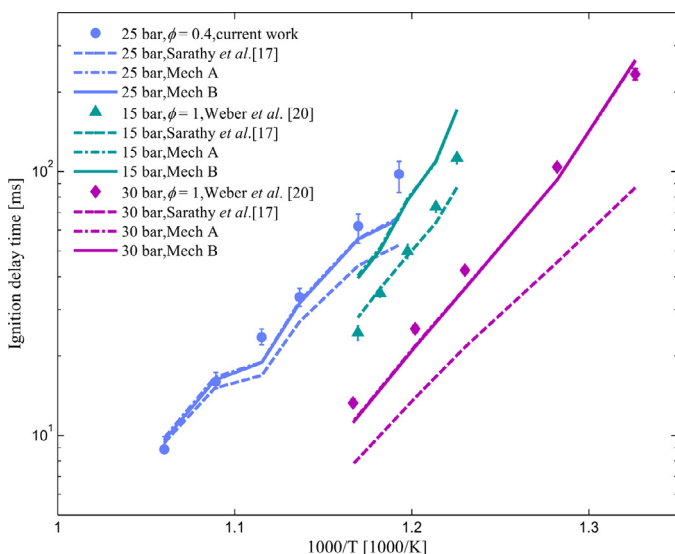


**Fig. 16.** Comparison between the measured intermediate species and the model predictions using variable volume model for the condition of  $P_{\text{eff}} = 25.3$  bar,  $T_{\text{eff}} = 905$  K,  $\chi(\text{iso-butanol}) = 0.8\%$ ,  $\chi(\text{O}_2) = 12.0\%$ ,  $\chi(\text{N}_2) = 55.9\%$ ,  $\chi(\text{Ar}) = 31.3\%$ .

**Table 1**

Reaction rate constants adopted in mechanism B. Units:  $\text{cm}^3/\text{mol/s/cal}$ . The rate coefficient expression is  $k = A (T/K)^b \exp(-E_a/RT)$ .

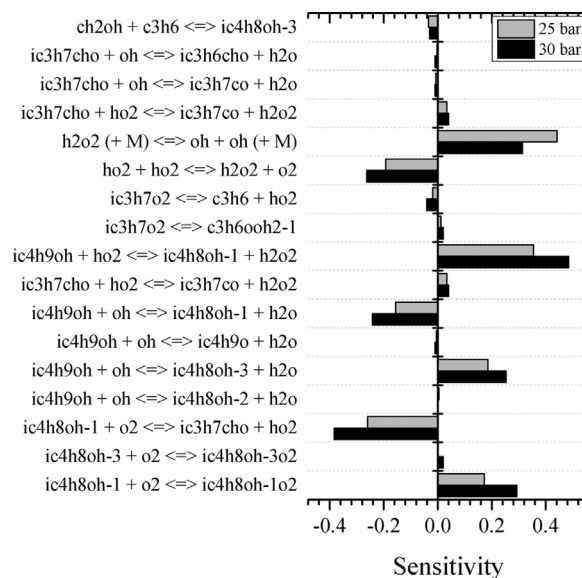
| No.        | Reaction                                                                                             | Pressure | A         | b      | E <sub>a</sub> | Refs. |
|------------|------------------------------------------------------------------------------------------------------|----------|-----------|--------|----------------|-------|
| 1          | $i\text{C}_4\text{H}_9\text{OH} + \text{OH} = i\text{C}_4\text{H}_8\text{OH-1} + \text{H}_2\text{O}$ |          | 3.792E+06 | 2.017  | -1205          | [30]  |
| 2          | $i\text{C}_4\text{H}_9\text{OH} + \text{OH} = i\text{C}_4\text{H}_8\text{OH-2} + \text{H}_2\text{O}$ |          | 1.207E+04 | 2.593  | -2641          | [30]  |
| 3          | $i\text{C}_4\text{H}_9\text{OH} + \text{OH} = i\text{C}_4\text{H}_8\text{OH-3} + \text{H}_2\text{O}$ |          | 5.127E+04 | 2.380  | -1320          | [30]  |
| 4          | $i\text{C}_4\text{H}_9\text{OH} + \text{OH} = i\text{C}_4\text{H}_9\text{O} + \text{H}_2\text{O}$    |          | 4.196E+02 | 2.917  | -364.54        | [30]  |
| 5          | $i\text{C}_4\text{H}_8\text{OH-1} = \text{CH}_3\text{CHCHOH} + \text{CH}_3$                          | (1 atm)  | 7.61E+31  | -6.15  | 32349          | [19]  |
| 6          | $i\text{C}_4\text{H}_8\text{OH-1} = i\text{C}_3\text{H}_7\text{CHO} + \text{H}$                      | (1 atm)  | 1.57E+31  | -6.195 | 32924.7        | [19]  |
| 7          | $i\text{C}_4\text{H}_8\text{OH-1} = i\text{C}_3\text{H}_6\text{CHOH} + \text{H}$                     | (1 atm)  | 9.08E+29  | -6.142 | 33602          | [19]  |
| 8          | $i\text{C}_4\text{H}_8\text{OH-2} = i\text{C}_4\text{H}_8 + \text{OH}$                               | (1 atm)  | 1.63E+16  | -1.970 | 15757          | [19]  |
| 9          | $i\text{C}_4\text{H}_8\text{OH-2} = i\text{C}_3\text{H}_6\text{CHOH} + \text{H}$                     | (1 atm)  | 4.75E+11  | -1.303 | 20383.5        | [19]  |
| 10         | $i\text{C}_4\text{H}_8\text{OH-2} = i\text{C}_3\text{H}_5\text{CH}_2\text{OH} + \text{H}$            | (1 atm)  | 5.47E+06  | -0.190 | 18752          | [19]  |
| Duplicated | $i\text{C}_4\text{H}_8\text{OH-2} = i\text{C}_3\text{H}_5\text{CH}_2\text{OH} + \text{H}$            | (1 atm)  | 8.76E+04  | 0.361  | 16827          | [19]  |
| 12         | $i\text{C}_4\text{H}_8\text{OH-3} = \text{C}_3\text{H}_6 + \text{CH}_2\text{OH}$                     | (1 atm)  | 1.67E+23  | -3.762 | 23321          | [19]  |
| 13         | $i\text{C}_4\text{H}_8\text{OH-3} = \text{C}_3\text{H}_5\text{OH} + \text{CH}_3$                     | (1 atm)  | 8.48E+21  | -3.566 | 25327          | [19]  |
| 14         | $i\text{C}_4\text{H}_8\text{OH-3} = i\text{C}_3\text{H}_5\text{CH}_2\text{OH} + \text{H}$            | (1 atm)  | 1.37E+18  | -3.104 | 26692          | [19]  |



**Fig. 17.** Comparison between the measured ignition delay time and the predictions using the mechanism developed by Sarathy et al. [17] as well as Mechanism A and Mechanism B.

calculating the thermodynamic properties of the fuel radicals. Instead of updating the rates of a single reaction. The rates of the whole class of  $\beta$ -scission reactions of the  $i\text{C}_4\text{H}_8\text{OH}$  radicals in Mechanism A were substituted by those computed by Cai et al. [19], yielding Mechanism B (provided in the Supplemental Material). Table 1 presents the rates modified in Mechanism B. Mechanism B predicts the ignition delay of 19.1 ms, within 3% of the prediction by Mechanism A at  $P_{\text{eff}} = 25.3$  bar,  $T_{\text{eff}} = 905$  K, and  $\chi$  (iso-butanol) = 0.8%. Figure 16 shows that Mechanism B slightly overpredicts the ethene concentrations than Mechanism A, but still significantly better than Sarathy mechanism. The Mechanism B significantly improves the predictions of iso-butene, comparing to Mechanism A.

Finally, both Mechanism A and Mechanism B were validated with the ignition delay time at 25 bar obtained in the current work as well as at 15 bar and 30 bar from Weber and Sung [20]. Figure 17 shows that Mechanism B predicts almost the same ignition delay with the Mechanism A, and it better reproduces the ignition delay times of iso-butanol for both the current study at 25 bar, 840–950 K within 33% of the measurements, and those of [20] at 30 bar, 754–857 K within 18% of the measurements. However Mechanism B overpredicts the ignition delay times of iso-butanol at 15 bar, 816–855 K by 50%, while the predictions by the Sarathy mechanism are within



**Fig. 18.** Linear brute force sensitivity analysis of the ignition delay with respect to the A factors of the listed reactions in the mechanism from Sarathy et al. [17]. The simulations were done at  $P = 25$  bar,  $T = 850$  K,  $\chi$  (iso-butanol) = 0.8%,  $\chi$  ( $\text{O}_2$ ) = 12.0%,  $\chi$  ( $\text{N}_2$ ) = 55.9%,  $\chi$  (Ar) = 31.3%, and  $P = 30$  bar,  $T = 850$  K,  $\chi$  (iso-butanol) = 3.38%,  $\chi$  ( $\text{O}_2$ ) = 20.30%,  $\chi$  ( $\text{N}_2$ ) = 76.52% using constant volume model. Positive quantities indicate the ignition delay is increased when the A factor is halved.

23% of the measurements. Noting that the ignition delay data from Weber and Sung [20] were reported with the pressure and temperature at the end of compression and the specific volume time history. All of the simulations were performed using the variable volume model.

A linear brute force sensitivity analysis was conducted to identify the key reactions leading to the changes of ignition delay time using the Mechanism B. Following the sensitivity analysis by Weber and Sung [20], the sensitivity coefficient is defined as the difference between the ignition delay when the A factor of each reaction is halved in both forward and reverse direction and the nominal ignition delay, normalized by the nominal ignition, as shown in Eq. (8):

$$\text{Sensitivity} = \frac{\tau(0.5A_i) - \tau(A_i)}{\tau(A_i)} \quad (8)$$

Therefore, negative sensitivity means that halving the A factor of a reaction decreases the ignition delay, and positive sensitivity indicates increase in ignition delay. The sensitivity analysis was performed



using the constant volume model. Figure 18 shows that the sensitivities at 25 bar, 850 K,  $\phi = 0.4$  are almost the same as those at 30 bar, 850 K,  $\phi = 1$ , and those at 30 bar, 750 K,  $\phi = 1$  in [20]. Based on the sensitivity analysis, both higher rates of  $\alpha$ -branch and lower rates of  $\gamma$ -branch of H-atom abstractions from *iso*-butanol by OH radical lead to longer ignition delay. Higher rates of  $iC_4H_8OH-1$  formation lead to longer ignition delay, which is because the dominant reaction  $iC_4H_8OH-1 + O_2 = iso\text{-butyraldehyde} + HO_2$  in the consumption of  $iC_4H_8OH-1$  radical essentially inhibits the low-temperature ignition behavior of *iso*-butanol, as indicated by Sarathy et al. [17]. Lower rates of  $iC_4H_8OH-3$  formation lead to longer ignition delay, which is because the dominant low-temperature chain branching pathway in the consumption of  $iC_4H_8OH-3$  radical speeds up the low temperature ignition behavior.

## 6. Conclusions

Using a rapid compression machine with a fast sampling system, time resolved histories of the intermediate species of methane, ethene, propene, *iso*-butene, *iso*-butyraldehyde and carbon monoxide, produced during the auto-ignition of *iso*-butanol were obtained. These data provide useful new insights into the low temperature mechanism of *iso*-butanol. Ignition delay times of *iso*-butanol at 25 bar, 840–950 K are also presented and compared with model predictions by the Sarathy mechanism, the Cai mechanism, and the Merchant mechanism. The Sarathy mechanism and the Merchant mechanism well predict the current ignition delay times. However, the Sarathy mechanism cannot reproduce the histories of ethene, *iso*-butene and *iso*-butyraldehyde, and the Sarathy mechanism was extensively investigated as initial mechanism.

Sensitivity and pathway analyses indicate that the histories of ethene and *iso*-butene are strongly affected by the H-atom abstractions of *iso*-butanol by the OH radical, and the competing ratios between the oxygen addition to fuel radicals and  $\beta$ -scission of fuel radicals. The Sarathy mechanism was then modified based on the recently published rate constants of *iso*-butanol + OH-abstractions and those of  $\beta$ -scissions from the hydroxybutyl radicals, leading to significant improvement in the accuracy of the mechanism in predicting the ethene and *iso*-butene concentrations. Furthermore, the modified mechanism also better predicts the present ignition delay times at 25 bar, 840–950 K obtained in this study and those at 30 bar, 754–857 K, reported by Weber and Sung [20]. However the modified mechanism still cannot well predicts current *iso*-butyraldehyde concentrations and the ignition delay times at 15 bar and at 816–855 K from Weber and Sung [20]. Further development on the low to intermediate temperature mechanism is required.

## Acknowledgements

The authors would like to acknowledge the valuable suggestions from Dr. S.M. Sarathy at KAUST in optimizing the mechanism. This

work was supported by the National Natural Science Foundation of China (Grant no. 51476086) and the State Key Laboratory of Automotive Safety and Energy.

## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2015.06.010.

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